

Logical Data Dictionary

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Document control

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Logical Data Dictionary - Titans InstaFood

Lineage: normalized from archive/Data-Dictionary-v1.0.pdf | Document version 4.0

Notation: PK = primary key, FK = foreign key

Entity relationship summary

Customer -< Order >- OrderHasItems >- Item

? ^

-? Payment ?

-? Delivery ?

Employee ----- (processes order)

Menu -< Item (catalog grouping)

Full diagram: SDLC/04-Design/diagrams/Logical-Entity-Relationship-Diagram-v4.0.png

1. Employee

Field	Type	Size	Null	Definition
EmployeeID	INT	-	N	PK. Format: YYYY + 5-digit seq (e.g. 202400001)
NamePrefix	VARCHAR	4	Y	Dr., Mr., Mrs., Miss., Ms.
FirstName	VARCHAR	30	N	
LastName	VARCHAR	30	N	
DateOfBirth	DATE	-	N	ISO date; *was Character(8) in v1.0 - corrected*
Position	VARCHAR	25	N	Job title
Phone	VARCHAR	15	N	Digits only, no punctuation stored
Email	VARCHAR	100	N	
Department	VARCHAR	50	Y	DR 2.5
DateOfHire	DATE	Y	DR 2.5	
EmploymentStatus	VARCHAR	20	Y	full-time, part-time, contract
WorkLocation	VARCHAR	50	Y	
EmergencyContactName	VARCHAR	60	Y	
EmergencyContactPhone	VARCHAR	15	Y	

2. Customer

Field	Type	Size	Null	Definition
CustomerID	INT	-	N	PK. MM + 8-digit seq (e.g. 130001225)
FirstName	VARCHAR	30	N	
LastName	VARCHAR	30	N	
DateOfBirth	DATE	Y		
StreetNumber	VARCHAR	9	N	USPS rules for fractions/alpha suffix
StreetPrefix	CHAR	2	Y	N,S,E,W,NE,NW,SE,SW
StreetName	VARCHAR	100	N	
StreetType	VARCHAR	5	N	USPS Pub. 28 abbreviation
StreetSuffix	CHAR	2	Y	
City	VARCHAR	30	N	Lookup from ZIP *Should*
State	CHAR	2	N	
ZipCode	CHAR	9	N	ZIP+4, no dash stored
Country	CHAR	2	Y	ISO country code
Phone	VARCHAR	15	N	*v1.0 mislabeled "Employee Phone" - corrected*
Email	VARCHAR	100	N	Unique index recommended
CreatedAt	TIMESTAMP	-	N	
LastActivityAt	TIMESTAMP	Y		DRRD5

3. Item

Field	Type	Size	Null	Definition
ItemID	BIGINT	-	N	PK. YYYYMMDD + 7-digit seq
Name	VARCHAR	30	N	
Description	TEXT	-	Y	
Price	DECIMAL	10,2	N	>= 0
Available	BOOLEAN	-	N	Default true
ImageUrl	VARCHAR	150	Y	HTTPS URL
CategoryCode	VARCHAR	20	N	Maps to FR24 categories
IngredientsInfo	TEXT	Y		Allergen text (DR 2.9)

4. Menu

| Field | Type | Size | Null | Definition |

|-----|-----|-----|-----|-----|

| MenuID | INT | - | N | PK. MM + 8-digit seq |

| MenuName | VARCHAR | 25 | N | |

| MenuDescription | TEXT | 100 | Y | |

| MenuImageUrl | VARCHAR | 150 | Y | *v1.0 size 10 - corrected* |

| ItemID | BIGINT | - | N | FK ? Item (M:N via junction if multiple items per menu row - normalize to MenuItem if needed) |

Note: If one menu contains many items, use junction table MenuItem(MenuID, ItemID).

5. Order

Field	Type	Size	Null	Definition
OrderID	INT	-	N	PK. YYYY + 5-digit seq
CustomerID	INT	-	N	FK ? Customer
EmployeeID	INT	Y	FK ? Employee (processor)	
OrderDate	DATE	-	N	
OrderTime	TIME	-	N	*v1.0 typed as Date - corrected*
OrderAmount	DECIMAL	10,2	N	System-calculated sum
Status	VARCHAR	20	N	e.g. Pending, Paid, Preparing, Ready, Completed, Cancelled
FulfillmentType	VARCHAR	10	N	PICKUP or DELIVERY (FR23)
SpecialInstructions	TEXT	Y		

6. Payment

Field	Type	Size	Null	Definition
PaymentID	INT	-	N	PK
OrderID	INT	-	N	FK ? Order (1:1 or 1:N if partial payments - default 1:1)
CustomerID	INT	-	N	FK ? Customer (payer)
PaymentAmount	DECIMAL	10,2	N	
PaymentMethod	VARCHAR	30	N	Card, wallet, etc.
PaymentStatus	VARCHAR	20	N	Successful, Failed
PaymentDate	TIMESTAMP	-	N	
GatewayTransactionId	VARCHAR	64	Y	Tokenized reference

7. Delivery

Field	Type	Size	Null	Definition
DeliveryID	INT	-	N	PK
OrderID	INT	-	N	FK ? Order
DeliveryStatus	VARCHAR	20	N	OutForDelivery, Delivered, Returned
DriverEmployeeID	INT	Y	FK ? Employee	
EstimatedDeliveryAt	TIMESTAMP	Y		
DeliveredAt	TIMESTAMP	Y		
StreetNumber ? Country	-	-	-	Copy of address at dispatch or FK to Customer address snapshot

8. OrderHasItems

Field	Type	Size	Null	Definition
OrderHasItemsID	INT	-	N	PK
OrderID	INT	-	N	FK ? Order
ItemID	BIGINT	-	N	FK ? Item
Quantity	INT	-	N	>= 1
UnitPrice	DECIMAL	10,2	N	Price at time of order
LineTotal	DECIMAL	10,2	N	Quantity × UnitPrice

Data quality rules

1. All monetary amounts: DECIMAL(10,2), banker's rounding at display.

2. No duplicate active email per customer.

3. OrderAmount = SUM(OrderHasItems.LineTotal) before tax/fees (tax *Should* be s

4. Soft-delete customers (FR3) vs. hard-delete per legal review.